

Capstone Project Report

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1. Problem Statement

The company is facing challenges in efficiently managing operations and maximising profitability. Specifically:

1. **Walmart Dataset:** The retail chain needs actionable insights and accurate forecasting models to improve store operations and match inventory with demand.
2. **Online Retail Dataset:** The company seeks to understand customer purchase behaviour and segment customers effectively to target marketing efforts.

2. Project Objective

1. **Walmart Dataset:**
 - To generate insights to improve store operations.
 - To develop forecasting models to predict weekly sales for the next 12 weeks for each store.
2. **Online Retail Dataset:**

- To analyse customer purchase behaviour.
- To segment customers using RFM analysis and clustering techniques.

3. Data Description

Walmart Dataset:

- **Columns:**
 - **Store:** Unique store identifier.
 - **Date:** Weekly sales date.
 - **Weekly_Sales:** Sales revenue for the week.
 - **Holiday_Flag:** Indicator of a holiday week.
 - **Temperature:** Temperature during the week.
 - **Fuel_Price:** Regional fuel price.
 - **CPI:** Consumer Price Index.
 - **Unemployment:** Regional unemployment rate.
- **Size:** 6,435 rows, 8 columns.

Online Retail Dataset:

- **Columns:**
 - **Invoice:** Unique invoice number.
 - **StockCode:** Product identifier.
 - **Description:** Product description.
 - **Quantity:** Number of products sold.
 - **InvoiceDate:** Date of purchase.
 - **Price:** Unit price of the product.
 - **CustomerID:** Unique customer identifier.
 - **Country:** Country of purchase.
- **Size:** 541,909 rows, 8 columns.

4. Data Preprocessing Steps and Inspiration

Walmart Dataset:

1. Converted `Date` to datetime format.
2. Handled missing values by removing rows with null values.
3. Added `Year`, `Month`, and `Week` features for time-based analysis.
4. Verified numerical consistency for columns like `Temperature` and `Weekly_Sales`.

Online Retail Dataset:

1. Dropped rows with missing `CustomerID` and `Description`.
2. Created `Total_Sales` as `Quantity * Price`.
3. Converted `InvoiceDate` to datetime format.
4. Removed canceled transactions (negative quantities).

5. Choosing the Algorithm for the Project

1. Walmart Dataset:

- **Prophet:** Suitable for time-series forecasting with multiple seasonality and holiday effects.

2. Online Retail Dataset:

- **RFM Analysis:** Measures Recency, Frequency, and Monetary value to rank customers.
- **K-Means Clustering:** Efficient for grouping customers based on RFM scores.

6. Motivation and Reasons for Choosing the Algorithm

1. Prophet:

- Handles time-series data with irregular intervals.
- Incorporates seasonality and holiday effects.

2. RFM Analysis and K-Means:

- RFM provides a clear understanding of customer behaviour.

- K-Means is efficient for clustering high-dimensional data like RFM scores.

7. Assumptions

1. Data is representative of typical sales and customer behaviours.
2. Holiday flags significantly influence sales trends in the Walmart dataset.
3. All transactions in the online retail dataset are valid after preprocessing.

8. Model Evaluation and Techniques

Walmart Dataset:

- **Evaluation Metrics:**
 - Root Mean Square Error (RMSE).
 - Mean Absolute Percentage Error (MAPE).

Online Retail Dataset:

- **Evaluation Techniques:**
 - RFM scores normalised using StandardScaler.
 - Cluster evaluation through visualisations and Silhouette scores.

9. Inferences

1. **Walmart Dataset:**
 - Holiday weeks show significant spikes in sales.
 - Temperature and unemployment have mild correlations with sales.
2. **Online Retail Dataset:**
 - High-value customers (low recency, high frequency, high monetary) contribute significantly to total sales.
 - Distinct customer segments can be targeted for promotions.

10. Future Possibilities

1. Include additional features (e.g., weather, promotions) for better forecasting.

2. Implement advanced clustering techniques like DBSCAN for customer segmentation.
3. Automate insights generation using dashboards in Tableau or Power BI.

11. Conclusion

This project provided actionable insights and robust models for sales forecasting and customer segmentation. It can guide inventory management, marketing strategies, and resource allocation for improved business outcomes.

12. References

1. Walmart dataset and project description.
2. Online retail dataset and project description.
3. Prophet documentation: <https://facebook.github.io/prophet/>
4. Scikit-learn documentation for K-Means clustering.